

Intro to HPC

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March 2025

This material is extended and adapted from
“How to start with spatial omics” module “Data Analysis” by Janick Mathys



HPC

High-performance
computing



High-performance computing

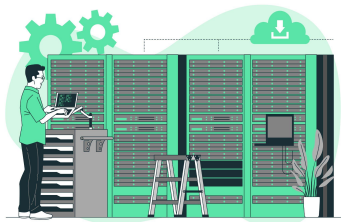
Computer



Server



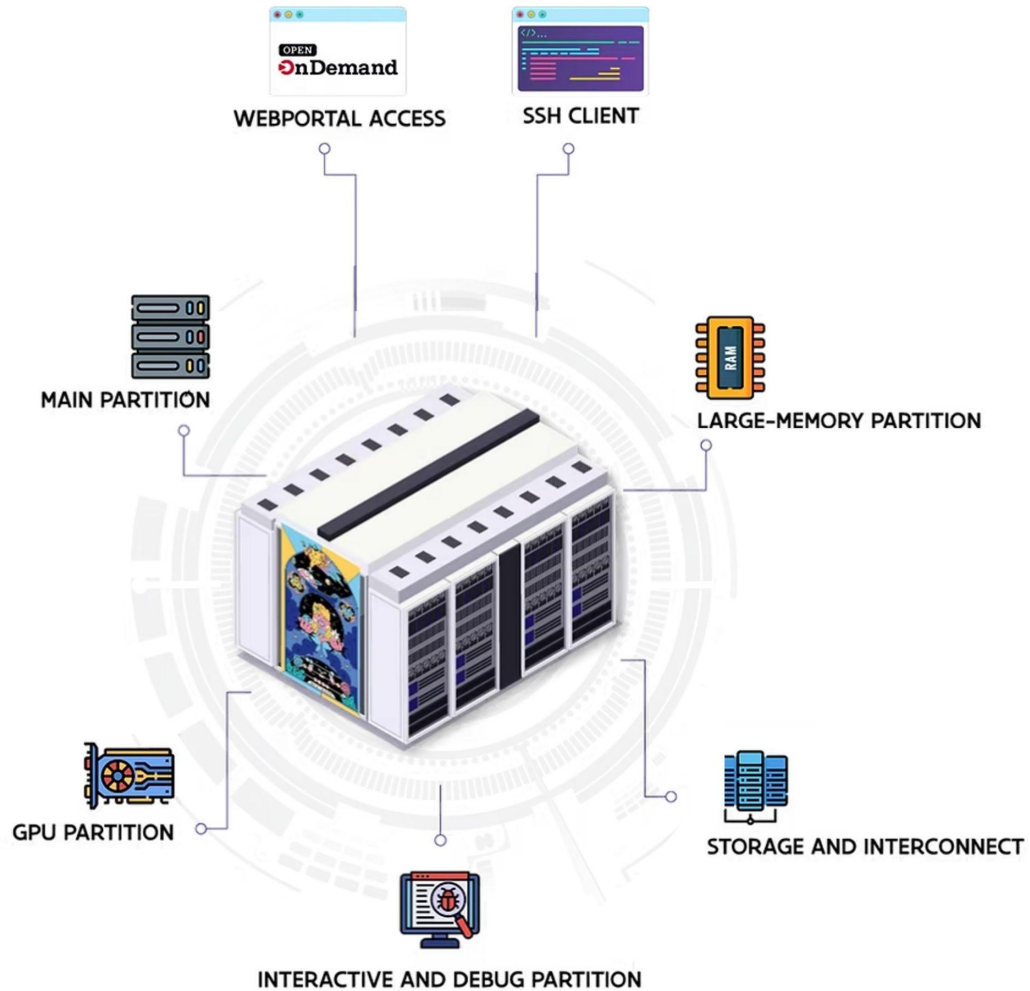
HPC



- Different resources available

- Nodes → parallel file system

- CPU with
 - Multi-processor interconnect
 - High memory
- GPU
 - Specialized processor
- Fast interconnection

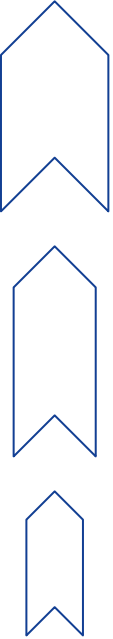
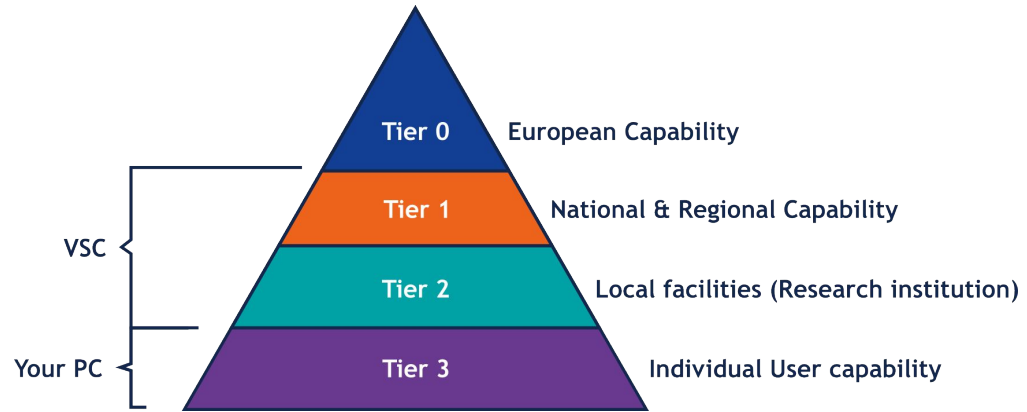


HPC

Tiers & Resources

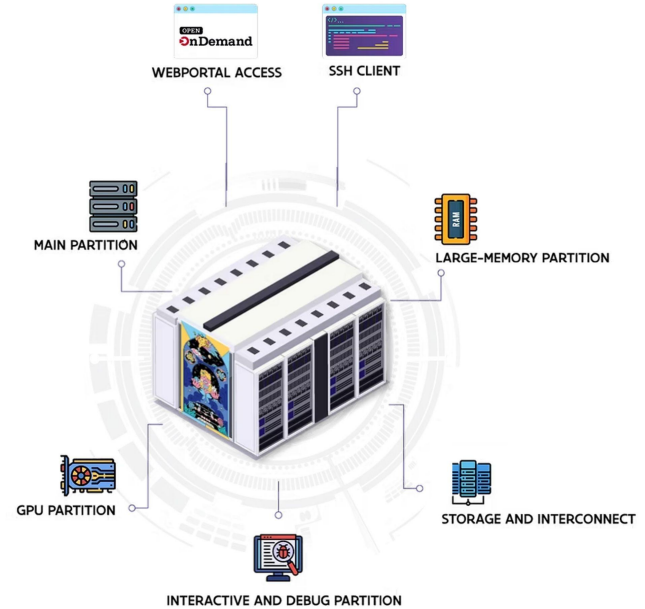
EU model: Infrastructure

- Tier-0
 - 6 in EU
 - Collaboration
- Tier-1 
 - Hortense (UGent)
- Tier-2 
 - Stevin (UGent)
 - Genius (KULeuven)
 - wICE (KULeuven)



VIB Data Core Compute Cluster

- A centralized Compute solution for VIB users
 - Similar infrastructure
 - Only for VIB employees
 - Not using the Tier system classification



HPC

File System



File System

Purpose & Maintenance

Capacity (GB)



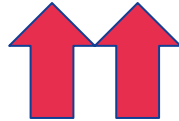
Home:



Data



Scratch



File System

Purpose & Maintenance

Capacity (GB) | Performance



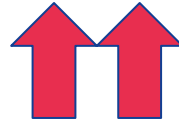
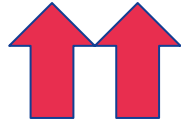
Home:



Data



Scratch



File System

Purpose & Maintenance

Capacity (GB) | Performance | Purpose

	Home:			▪ Config and user preferences ▪ No job execution ▪ No data
	Data			
	Scratch			

File System

Purpose & Maintenance

Capacity (GB) | Performance | Purpose



Home:



- Config and user preferences
- No job execution
- No data



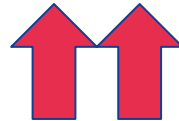
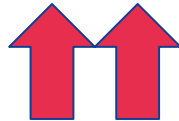
Data



- Long-term storage
- No job execution



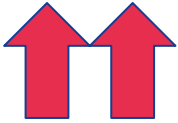
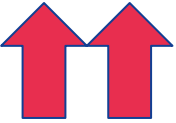
Scratch



File System

Purpose & Maintenance

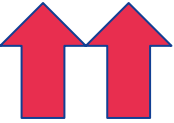
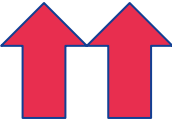
Capacity (GB) | Performance | Purpose

	Home:			▪ Config and user preferences ▪ No job execution ▪ No data
	Data			▪ Long-term storage ▪ No job execution
	Scratch			▪ Short-term storage ▪ Job executed ▪ Not accessible in other VSCs

File System

Purpose & Maintenance

Capacity (GB) | Performance | Purpose

	Home:			▪ Config and user preferences ▪ No job execution ▪ No data
	Data			▪ Long-term storage ▪ No job execution
	Scratch			▪ Short-term storage ▪ Job executed ▪ Not accessible in other VSCs

Copy
before run

File System

Different systems



\$VSC_HOME



\$VSC_DATA



\$VSC_SCRATCH



Other Scratch

* \$HOME = \$VCS_SCRATCH

** Project based

File System

Different systems

**
 Tier1
UNIVERSITEIT
GENT

|  Tier2
UNIVERSITEIT
GENT

**
 Tier 2

	<code>\$VSC_HOME</code>	* <code>\$HOME</code>	3 GB	3 GB
	<code>\$VSC_DATA</code>	Variable	25 GB	75 GB
	<code>\$VSC_SCRATCH</code>	3 GB	25 GB	500 GB
	Other Scratch	Variable	X	up to 600 GB

`$VSC_SCRATCH_PROJECTS_BASE`

`$VSC_SCRATCH_NODE`

File System

Different systems

** Project based

**  Tier1 UNIVERSITEIT GENT		 Tier2 UNIVERSITEIT GENT		**  Tier 2		**  VIB Data Core compute
--	--	---	--	--	--	--

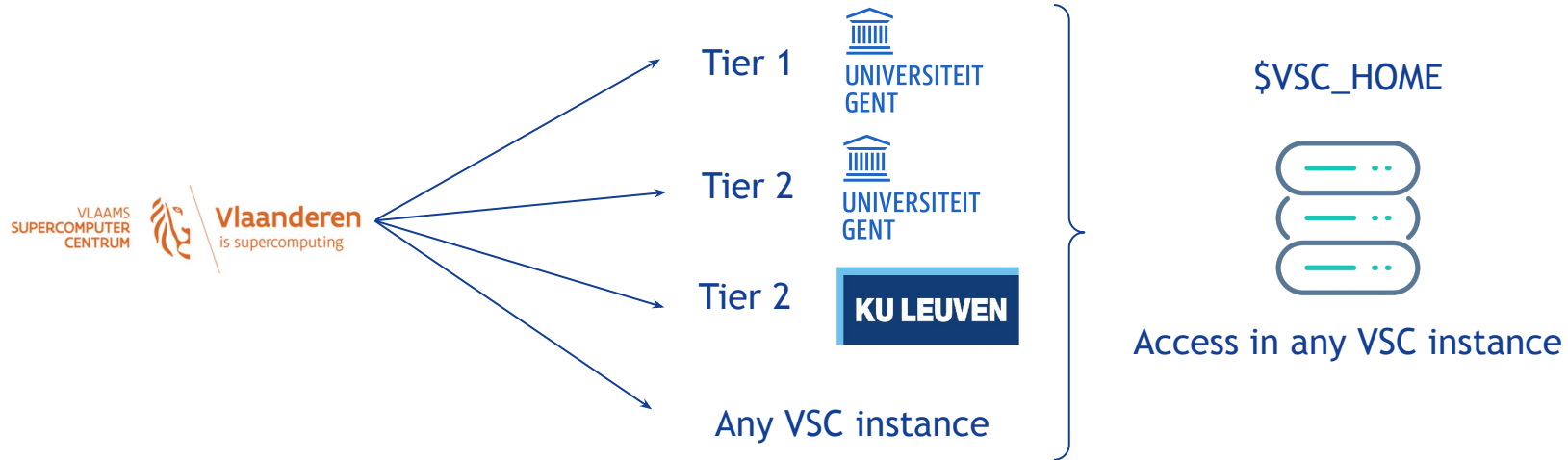
	\$VSC_HOME	* \$HOME	3 GB	3 GB	\$HOME	10 GB
	\$VSC_DATA	Variable	25 GB	75 GB	path/to/	Variable
	\$VSC_SCRATCH	3 GB	25 GB	500 GB	X	X
	Other Scratch	Variable	X	up to 600 GB	\$SCRATCHDIR	Variable

\$VSC_SCRATCH_PROJECTS_BASE

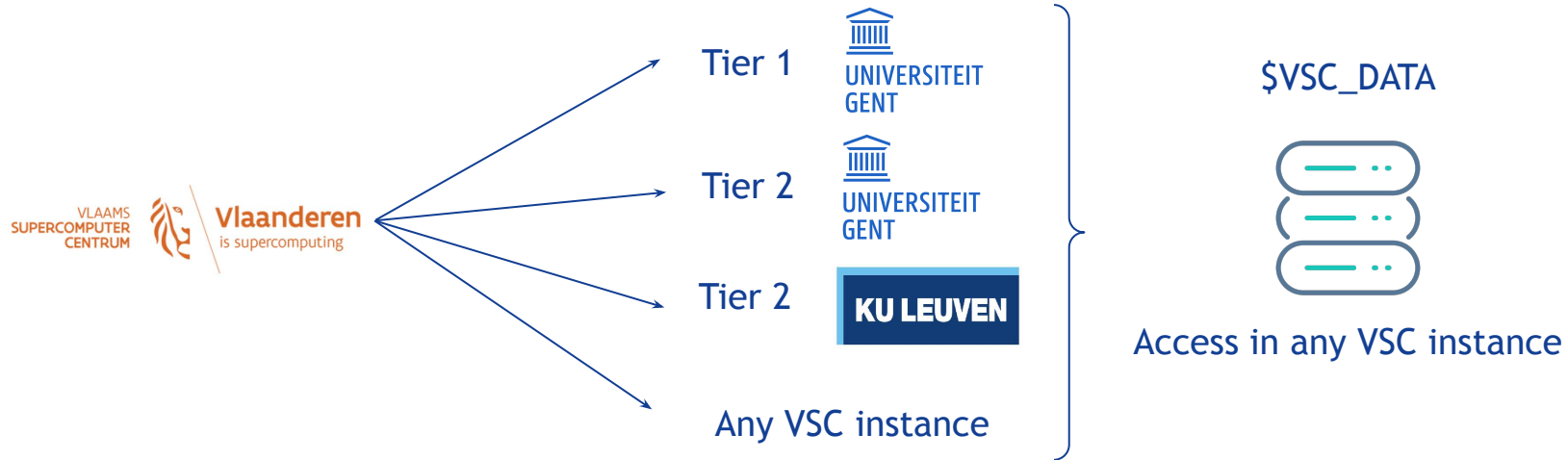
\$VSC_SCRATCH_NODE

Local to Node

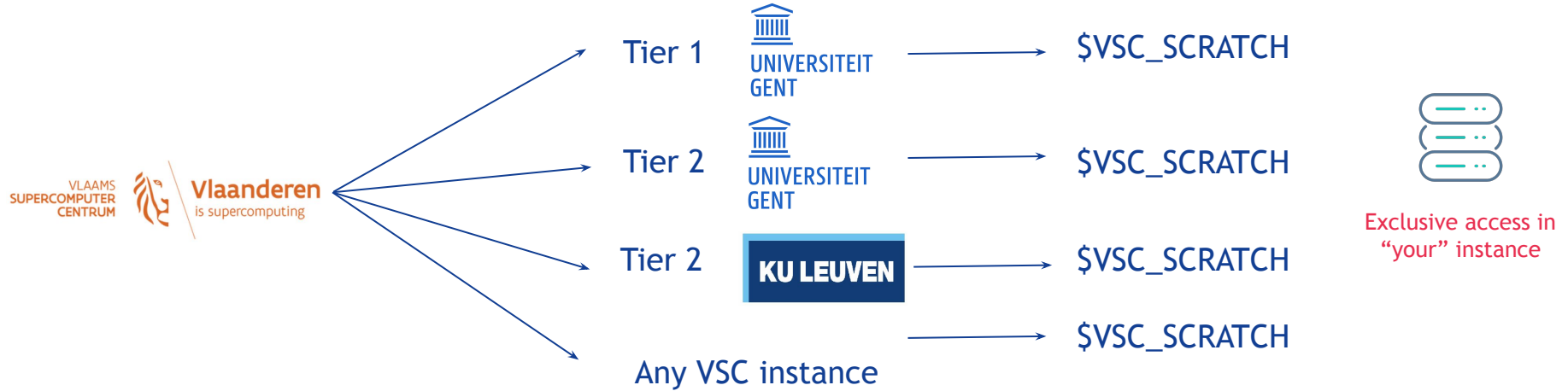
Accessible from:



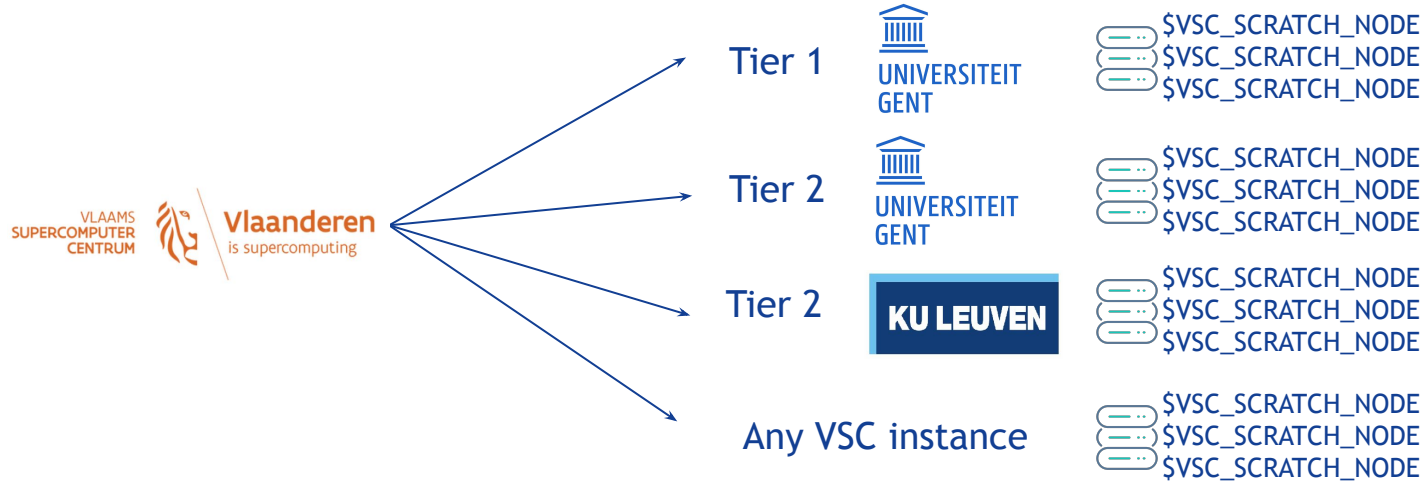
Accessible from:



Accessible from:



Accessible from:



Exclusive of
each node

Similar for  **VIB** Data Core compute

HPC

Node offer

Nodes in HPC

- Variable resources
 - CPUs
 - GPUs
 - Memory
- Debug interactive node
- Each node has a name



Nodes in HPC

- Variable resources
 - CPUs
 - GPUs
 - Memory
- Debug interactive node
- Each node has a name

} More resources == Less priority

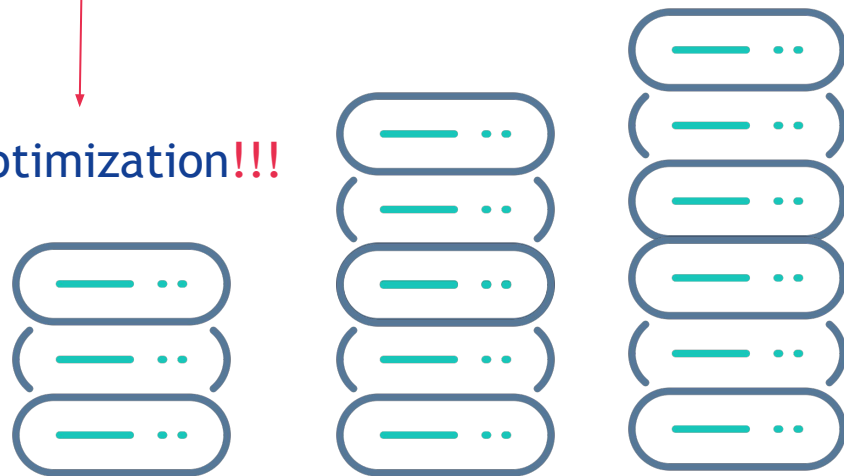


Nodes in HPC

- Variable resources
 - CPUs
 - GPUs
 - Memory
- Debug interactive node
- Each node has a name

} More resources == Less priority

Optimization!!!



Nodes in HPC

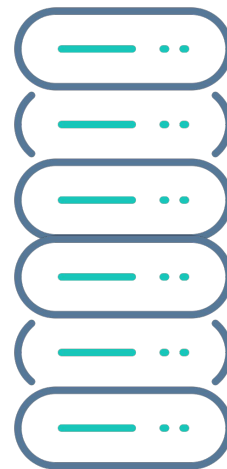
- Variable resources
 - CPUs
 - GPUs
 - Memory

Look into *log file*

jobname.logID



Optimization!!!



Nodes and names



debug_28C_56T_750GB

VIB



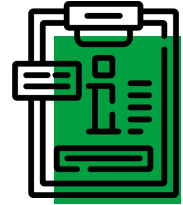
donphan

Tier 2 UGent



debug_rome

Tier 1 UGent



interactive
gpu_a100_debug

Tier 2 KULeuven

Nodes and names



debug_28C_56T_750GB

VIB



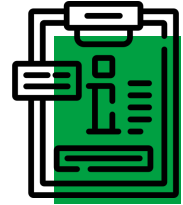
donphan

Tier 2 UGent



debug_rome

Tier 1 UGent



interactive
gpu_a100_debug

Tier 2 KULeuven

HPC

Connect to OnDemand

OnDemand: Friendly interface

<https://website.be>



Files



Jobs



Clusters



Interactive Apps



Open OnDemand provides an integrated, single access point for all your HPC resources.



Active Jobs
System Installed App



Visual Studio Code
System Installed App



ParaView
System Installed App

System status:

PARTITION	AVAIL	TIMELIMIT	NODES	STATE	MODELIST
debug_28C_56T_750GB	up	14-00:00:0	1	mix	p-vibcompute-1
gpu_a100_48C_96T_512GB	up	14-00:00:0	1	mix	p-vibcompute-gpu-1
gpu_h100_64C_128T_2TB	up	14-00:00:0	4	mix	p-computegpu-[2-5]
gpu_h100_64C_128T_4TB	up	14-00:00:0	1	mix	p-computegpu-6
gpu_h100_64C_128T_4TB	up	14-00:00:0	1	idle	p-computegpu-7
gpu_h100_64C_128T_2TB_co_pi	up	14-00:00:0	4	mix	p-computegpu-[2-5]
gpu_h100_64C_128T_4TB_co_pi	up	14-00:00:0	1	mix	p-computegpu-6
gpu_h100_64C_128T_4TB_co_pi	up	14-00:00:0	1	idle	p-computegpu-7
gpu_140s_64C_128T_1TB	up	14-00:00:0	1	idle	p-computegpu-8
hmem_128C_256T_2TB	up	14-00:00:0	1	mix	p-computehmem-1
hmem_128C_256T_2TB	up	14-00:00:0	1	idle	p-computehmem-2
gp_64C_128T_512GB*	up	14-00:00:0	4	idle	p-computegp-[1-4]

VIB

<https://compute.vib.be>

UGent Tier-2

<https://login.hpc.ugent.be>

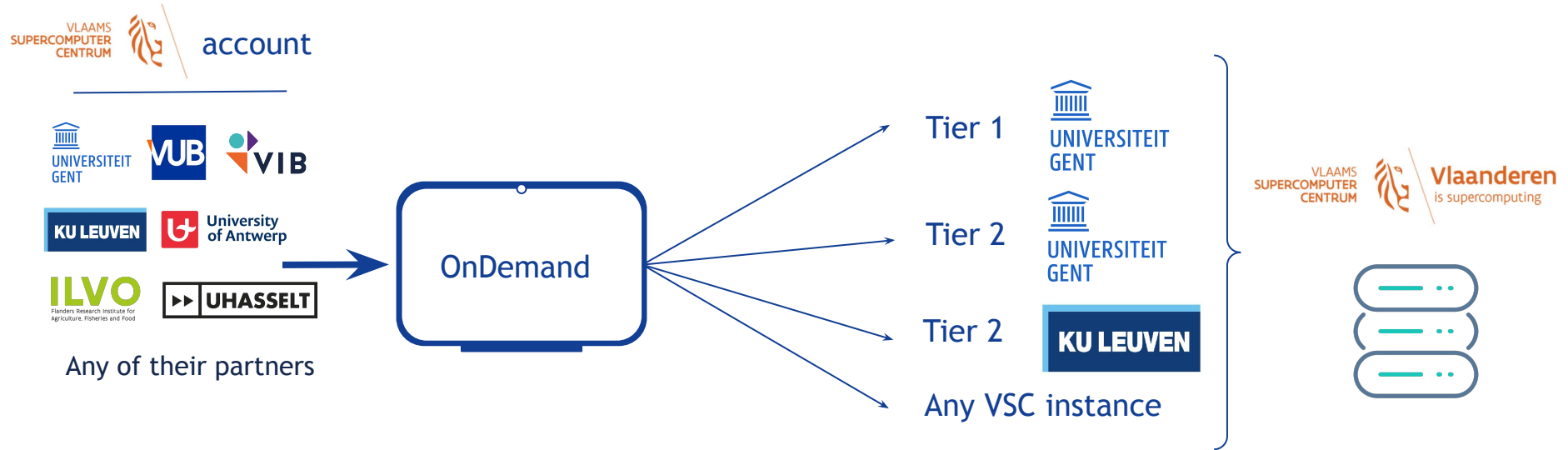
UGent Tier-1

<https://tier1.hpc.ugent.be/>

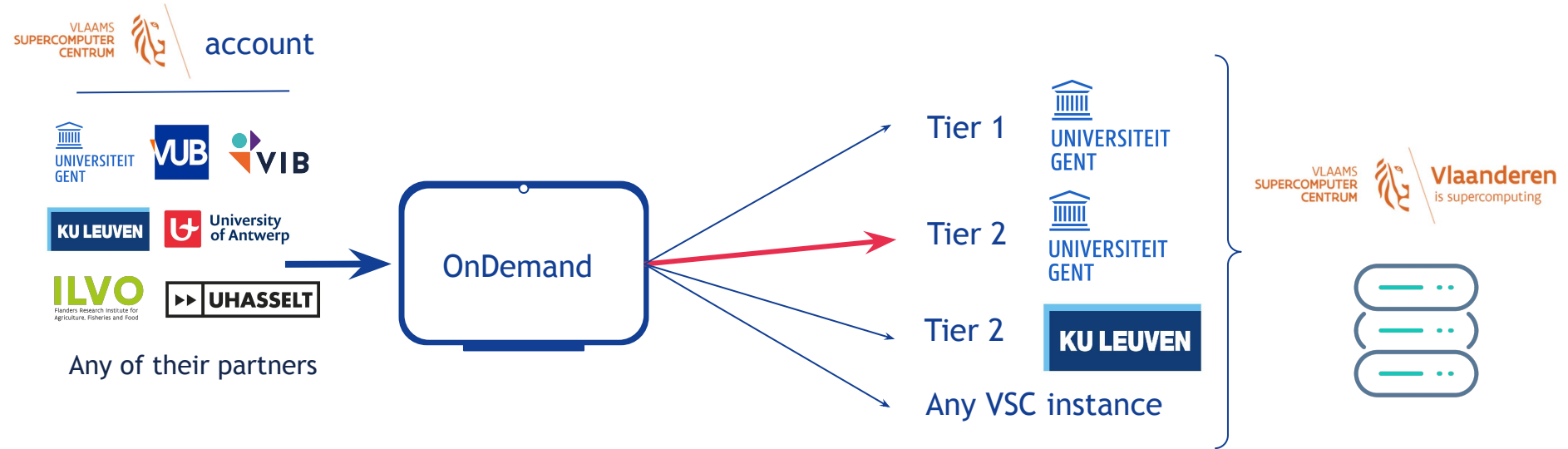
KULeuven Tier-2

<https://ondemand.hpc.kuleuven.be/>

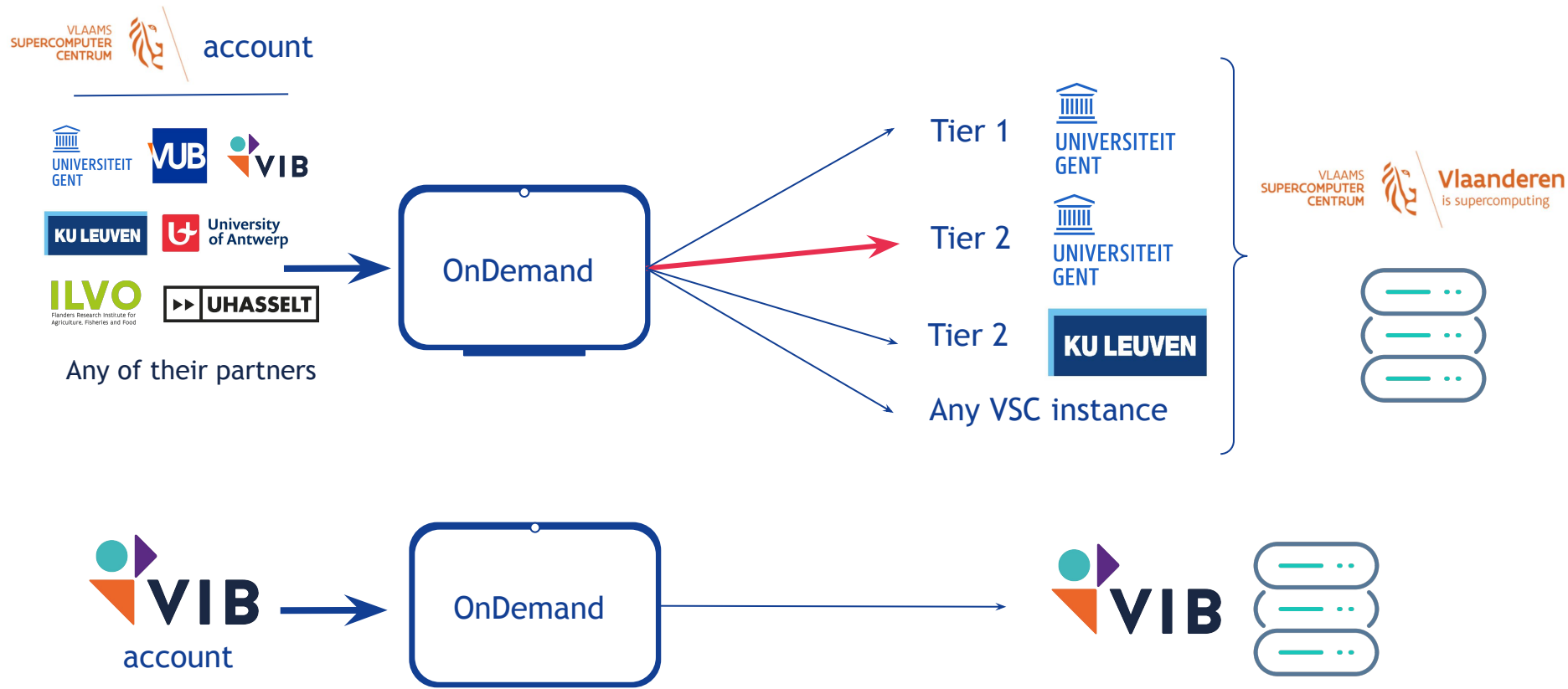
Where can you connect ?



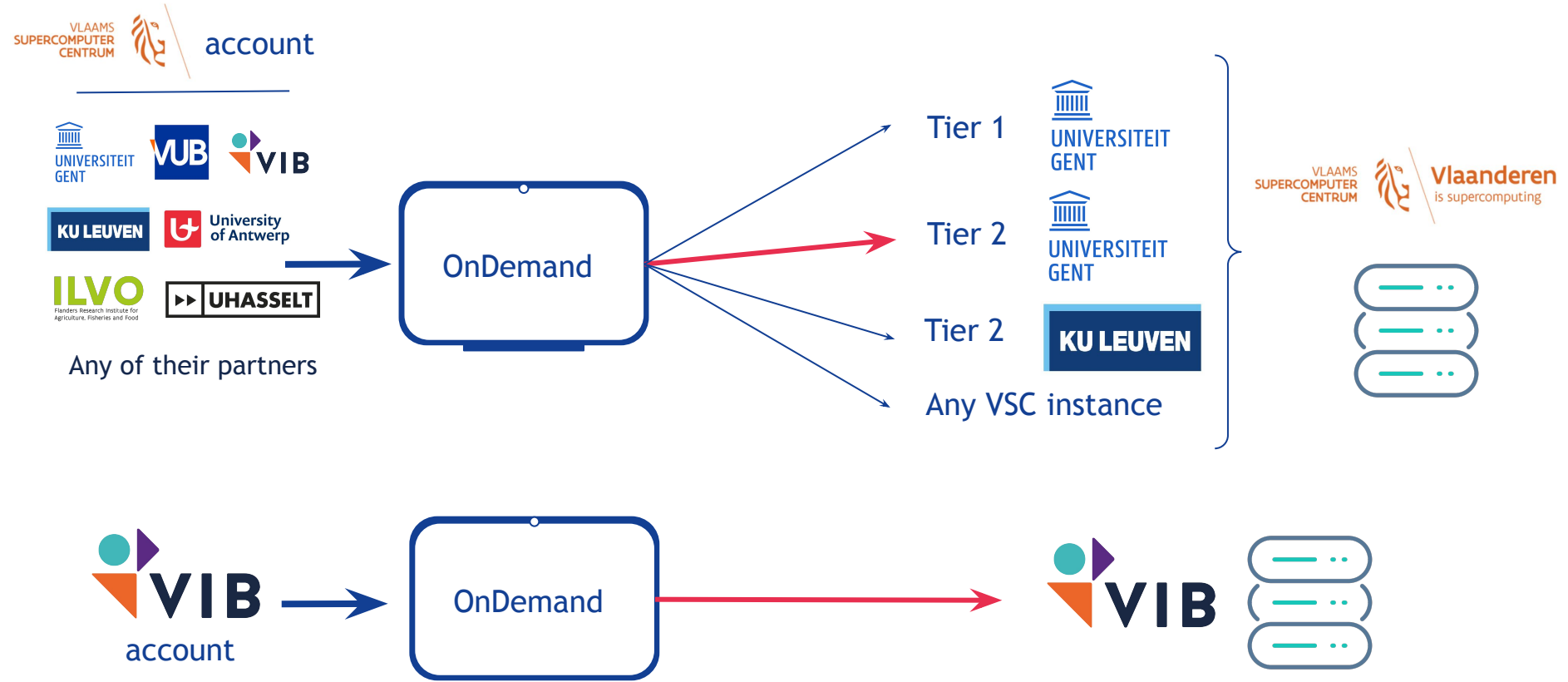
Where can you connect ?



Where can you connect ?



Where can you connect ?



Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions

KU Leuven VSC Authentication

Please choose a login provider.



KU Leuven users: VSC account



Partner organizations: VSC account

Or enter your temporary credentials: Username Password


science meets life

CENTRAL LOGIN

Authenticate with one of these centers. The very first login should be done with VIB credentials.

VIB	PSB
CMB	IRC



In order to proceed, you have to select your Home Organization from the list below.

VIB

- ArteveldeHogeschool
- EhB
- Flanders Make VZW - Azure AD SSO
- Flanders Marine Institute (VLIZ)
- Ghent University
- Hasselt University
- Hogent
- Howest
- Instituut voor Landbouw-, Visserij- en Voedingsonderzoek
- Instituut voor Tropische Geneeskunde Identity Provider
- KU Leuven Association
- Karel de Grote Hogeschool
- Royal Belgian Institute of Natural Sciences
- Royal Meteorological Institute of Belgium
- UNIVERSITAIR ZIEKENHUIS BRUSSEL
- UZ Gent
- Universiteit Antwerpen
- VIB**
- VITO

Let's connect

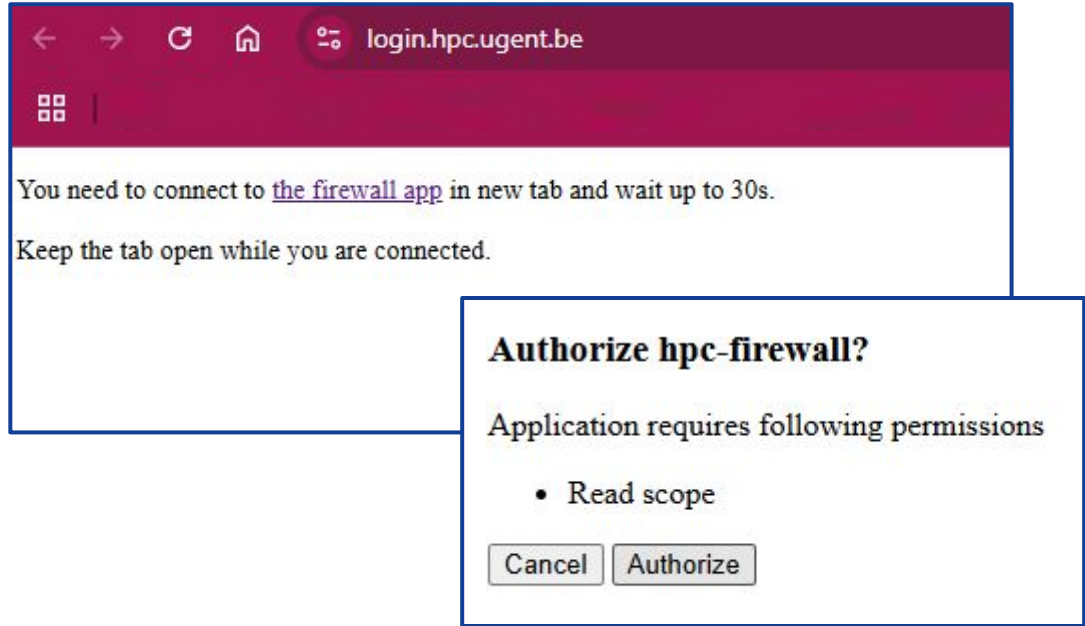
Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access



HPC

Connect cmd-line

Create an SSH key

- Check existing keys

```
ls ~/.ssh
```

- Generate a new key **just** for VSC

```
ssh-keygen -t ed25519 -C "your_email@example.com" -f ~/.ssh/ed25519_vsc
```

P.S.: Choose a strong password

- Check your new key

```
ls ~/.ssh
```

Inform the key to VSC

- Access your account

<https://account.vscentrum.be/django//>



→ Add public key

i You can reveal the .ssh folder on macOS and Linux by:

- on MacOS: press `cmd+shift+.` in Finder
- on linux: press `ctrl+h`

Public key file

→ Choose File No file chosen

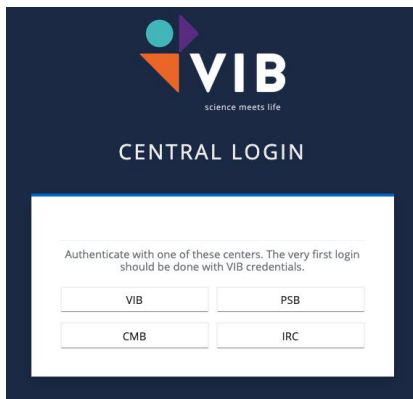
→ Upload extra public key

→ Update

It can take some time to be active

Authentication for VIB Data Core Compute Cluster

- Guide on docs.datacore.vib.be/entrypoints/command-line-access/
- Smallstep CLI leverages VIB SSO for a secure connection



- Generates certificate that is valid for 16 hours

Access from WSL/LINUX terminal

Log in

- VSC

```
ssh vsc00000@login.hpc.kuleuven.be
```

```
ssh vsc00000@login.hpc.ugent.be
```

- VIB

```
ssh -p 2022 firstname.lastname@compute.vib.be
```

We will try later

We need to wait the key to be authorized

On demand

Working &
checking resources

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP

Interactive Apps ▾

Servers

-  JupyterLab
-  Local LLM chat (beta)
-  RStudio
-  Visual Studio Code

 Shell

Interactive Apps ▾

Servers

-  Gnuplot
-  Interactive Shell
-  JupyterLab
-  NVIDIA RAPIDS
-  ParaView
-  RStudio Server
-  TensorBoard
-  Visual Studio Code

Interactive Apps ▾

Desktops

-  Bioimage ANalysis Desktop
-  Cluster Desktop
-  Neurodesk

Servers

-  Jupyter Lab
-  Jupyter Notebook
-  RStudio server
-  Shell (tmux)
-  VS Code Tunnel

Testing

-  Cluster desktop v2
-  Code Server
-  Neurodesk v2
-  Transcribe

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
 - Define resources

Home / My Interactive Sessions / VS Code Tunnel

Interactive Apps

Desktops

Bioimage ANalysis Desktop

Cluster Desktop

Neurodesk

Servers

Jupyter Lab

Jupyter Notebook

RStudio server

Shell (tmux)

VS Code Tunnel

Testing

Cluster desktop v2

Code Server

Neurodesk v2

Transcribe

This app will launch a [redacted] for remote development. More info can be found in the [redacted] documentation. (Depending on the version, the tunnel URL might also be used directly to connect to a VS Code server using a browser; but we recommend the [redacted] app for that).

An UGent user might get an error message `You do not have access to this` during the `Microsoft Authentication` step. If this is the case then, you might have to whitelist yourself at `UGent selfservice deviceCodeAuth`. You have to do it only once and you can provide [redacted] as motivation.

Cluster

[donphan (interactive/debug)]

Cluster

[donphan]

Time (hours)

[4 hours]

Number of cores (and default memory) per node

[4 cores (13.3 GiB mem)]

☐ I would like to receive an email when the session starts

☐ Show advanced options

Launch

* The VS Code Tunnel session data for this session can be accessed under the `data root directory`.

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
 - Define resources

Home / My Interactive Sessions / VS Code Tunnel

Interactive Apps

Desktops

- Bioimage ANalysis Desktop
- Cluster Desktop
- Neurodesk

Servers

- Jupyter Lab
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- Shell (tmux)

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Cluster

[donphan (interactive/debug)]

Cluster

[donphan]

[redacted]

[redacted]

[redacted]

starts

[redacted]

can be accessed under the data

[Credit account]
(<https://docs.vscentrum.be/leuven/credits.html>)

[zx_accont_project]

Reservation (optional)

[local_training]

Name of an existing reservation in which the job should run

KU LEUVEN

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
 - Define resources

Launch → wait ...

The screenshot displays the 'VS Code Tunnel' interface. On the left, a sidebar lists 'Interactive Apps' under 'Desktops' (Bioimage ANalysis Desktop, Cluster Desktop, Neurodesk) and 'Servers' (Jupyter Lab, Jupyter Notebook, RStudio server). The main area shows instructions for launching the app and a dropdown menu for selecting a cluster, currently set to 'donphan (interactive/debug)'. Below this, a status box indicates 'Your session is currently starting... Please be patient as this process can take a few minutes.' At the bottom, a detailed session card for 'Visual Studio Code (57988804)' shows it is running on 1 node and 1 core. It includes the host address 'r23i27n01.genius.hpc.kuleuven.be', creation time '2025-03-27 13:53:44 CET', time remaining '59 minutes', and session ID '83b34b81-4530-4d92-85ca-4b63f70e8d99'. A 'Connect to Visual Studio Code' button is at the bottom of the card. Red arrows point to the cluster dropdown, the status box, and the session card.

Let's connect

login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfmdZDTbm9

```
[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
[vsc47776@node4013 ~]$ tmux -S /tmp/J20159863.sock wait-for -S CUSTOMCHANNEL
[vsc47776@node4013 ~]$ reset
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ echo "Copy the device code, open the url and paste it"
Copy the device code, open the url and paste it
[vsc47776@node4013 ~]$ echo

[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
```



Let's connect

```
login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfdZDTbm9

[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
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[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code LYWC6GL6Y to authenticate.
LYWC6GL6Y[2025-03-27 14:25:13] info Creating tunnel with the name: vsc-vsc47776-donphan
```

Let's connect

```
login.hpc.ugent.be/node/node4013.donphan.os/13465/?arg=RxxpXAYJGAcfmdZDTbm9

[vsc47776@node4013 ~]$ export VSCODE_CLI_USE_FILE_KEYCHAIN=1
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[vsc47776@node4013 ~]$ code tunnel --accept-server-license-terms --name ${TUNNEL_NAME:0:20}
*
* Visual Studio Code Server
*
* By using the software, you agree to
* the Visual Studio Code Server License Terms (https://aka.ms/vscode-server-license) and
* the Microsoft Privacy Statement (https://privacy.microsoft.com/en-US/privacystatement).
*
✓ How would you like to log in to Visual Studio Code? · Microsoft Account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code LYWC6GL6Y to authenticate.
LYWC6GL6Y[2025-03-27 14:25:13] info Creating tunnel with the name: vsc-vsc47776-donphan

Open this link in your browser https://vscode.dev/tunnel/vsc-vsc47776-donphan/kyukon/home/gent/477/vsc47776
```

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
 - Define resources
- Choose the Shell
 - Start in \$HOME

Utilities ▾

Login node

>_ Quick Shell Access

Clusters ▾

>_ Login Server Shell Access

Clusters ▾

>_ RHEL9 Login node Shell Access

----- Connects in \$HOME -----

Interactive Apps ▾

Servers

-  JupyterLab
-  Local LLM chat (beta)
-  RStudio
-  Visual Studio Code

■ Shell

Interactive Apps ▾

Servers

-  Gnuplot
-  Interactive Shell
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-  NVIDIA RAPIDS
-  ParaView
-  RStudio Server
-  TensorBoard
-  Visual Studio Code

Interactive Apps ▾

Desktops

-  Bioimage ANalysis Desktop
-  Cluster Desktop
-  Neurodesk

Servers

-  Jupyter Lab
-  Jupyter Notebook
-  RStudio server
-  Shell (tmux)
-  VS Code Tunnel

Testing

-  Cluster desktop v2
-  Code Server
-  Neurodesk v2
-  Transcribe

Let's connect

Chose one or both:

[VIB](#)

[UGent Tier-2](#)

Steps to follow:

- Choose your institutions
- Authorize access
- Choose the APP
 - Define resources
- Choose the Shell
 - Start in \$HOME
 - Move to cluster
 - Module swap <cluster>

```
← → ↺ ↻ compute.vib.be/pun/sys/shell/ssh/vm-vibcomputelogs.vib.local
❏ |
Host: vm-vibcomputelogs.vib.local

PARTITION      AVAIL  TIMELIMIT  NODES  STATE MODELIST
debug_28C_56T_750GB  up 14-00:00:0  1  mix  p-vibcompute-1
gpu_a100_48C_96T_512GB  up 14-00:00:0  1  idle  p-vibcompute-gpu-1
gpu_h100_64C_128T_2TB  up 14-00:00:0  2  mix  p-computegpu-[2-3]
gpu_h100_64C_128T_2TB  up 14-00:00:0  2  idle  p-computegpu-[4-5]
gpu_h100_64C_128T_4TB  up 14-00:00:0  1  mix  p-computegpu-6
gpu_h100_64C_128T_4TB  up 14-00:00:0  1  idle  p-computegpu-7
gpu_h100_64C_128T_2TB_co_pi  up 14-00:00:0  2  mix  p-computegpu-[2-3]
gpu_h100_64C_128T_2TB_co_pi  up 14-00:00:0  2  idle  p-computegpu-[4-5]
gpu_h100_64C_128T_4TB_co_pi  up 14-00:00:0  1  mix  p-computegpu-6
gpu_h100_64C_128T_4TB_co_pi  up 14-00:00:0  1  idle  p-computegpu-7
gpu_l40s_64C_128T_1TB  up 14-00:00:0  1  mix  p-computegpu-8
hmem_128C_256T_2TB  up 14-00:00:0  2  mix  p-computehmem-[1-2]
gp_64C_128T_512GB*  up 14-00:00:0  4  idle  p-computegp-[1-4]
Rocky Linux 9.5 (Blue Onyx)
The VIB cluster has 6 partitions
- gpu_a100_48C_96T_512GB has 1 node with 4 nvidia A100 PCIe gpu's with 80GB of HBM2 memory,
  it also has 2 (ice lake) cpu's with 24 cores or 48 threads and 512GB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for highly parallel gpu specific jobs.
- gpu_h100_64C_128T_2TB has 4 nodes with 4 nvidia H100 SXM NVLink gpu's with 80GB of HBM3 memory,
  it also has 2 (sapphire rapids) cpu's with 32 cores or 64 threads and 2TB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for highly parallel gpu specific jobs.
- gpu_h100_64C_128T_4TB has 2 nodes with 4 nvidia H100 SXM NVLink gpu's with 80GB of HBM3 memory,
  it also has 2 (sapphire rapids) cpu's with 32 cores or 64 threads and 4TB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for highly parallel gpu specific jobs.
- gpu_l40s_64C_128T_1TB has 1 node with 4 nvidia L40s gpu's with 48GB of GDDR6 memory,
  it also has 2 (emerald rapids) cpu's with 32 cores or 64 threads and 1TB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for highly parallel gpu specific jobs.
- hmem_128C_256T_2TB has 2 nodes with 4 (sapphire rapids) cpu's with 32 cores and 2TB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for jobs requiring a lot of memory.
- gp_64C_128T_512GB has 4 nodes with 2 (ice lake) cpu's with 32 cores or 64 threads and 512GB of RAM.
  Local NVMe storage is available with $SCRATCHDIR. Use it for every non-debug, non-high mem, non-gpu job.
  This is the default partition.
- debug_28C_56T_750GB is a single node with 2 (haswell) cpu's with 14 cores or 28 threads and 768GB of RAM.
  Use it to debug your pipeline with a small dataset.
Default walltime unless specified is 1h.
Documentation can be found on https://docs.datacore.vib.be.
Last login: Thu Mar 27 14:31:01 2025 from 192.168.224.2
[bruna.piereckmoura@vm-vibcomputelogs ~]$
```

HPC

Tools available



Terminal Tools

Search for tools:

```
$ module avail <name>  
$ module av <name>  
$ module spider <name>
```

- D = default
- E = extension

```
[vsc47776@node4016 ~]$ module avail blast
```

```
----- /apps/gent/RHEL9/cascadelake-ib/modules/all -----  
BLAST+/2.14.1-gompi-2023a    BLAST+/2.16.0-gompi-2023b    BLAST+/2.16.0-gompi-2024a (D)    crb-blast/0.6.9-GCC-12.3.0  
  
----- This is a list of module extensions. Use "module --nx avail ..." to not show extensions. -----  
crb-blast (E)
```

These extensions cannot be loaded directly, use "module spider extension_name" for more information.

Where:

D: Default Module

E: Extension that is provided by another module

If you need software that is not listed, request it via <https://www.ugent.be/hpc/en/support/software-installation-request>

Terminal Tools

Search for tools:

```
$ module avail <name>  
$ module av <name>  
$ module spider <name>
```

- D = default
- E = extension

Load and use

```
$ module load <name>
```

```
[vsc47776@node4016 ~]$ module load BLAST+/2.16.0-gompi-2024a  
[vsc47776@node4016 ~]$ _
```

```
[vsc47776@node4016 ~]$ module avail blast
```

```
----- /apps/gent/RHEL9/cascadelake-ib/modules/all -----  
BLAST+/2.14.1-gompi-2023a    BLAST+/2.16.0-gompi-2023b    BLAST+/2.16.0-gompi-2024a (D)    crb-blast/0.6.9-GCC-12.3.0  
  
----- This is a list of module extensions. Use "module --nx avail ..." to not show extensions. -----  
crb-blast (E)  
  
These extensions cannot be loaded directly, use "module spider extension_name" for more information.  
  
Where:  
D: Default Module  
E: Extension that is provided by another module  
  
If you need software that is not listed, request it via https://www.ugent.be/hpc/en/support/software-installation-request
```

Terminal Tools

Check loaded modules:

```
$ module list <name>
```

- Modules will be available
 - in the node
 - during time requested

Unload modules

```
$ module unload <name>  
$ module purge
```



Terminal Tools

Check loaded modules:

```
$ module list <name>
```

- Modules will be available
 - in the node
 - during time requested

Unload modules

```
$ module unload <name>  
$ module purge
```

Some resources are promptly available

If you want to do a request, check if a module exists to facilitate:

<https://docs.easybuild.io/api/easybuild/tools/modules/>



Terminal Tools

Automatically available:

- Example:
 - python
 - singularity
 - apptainer
 - perls
 - sqlite
 - git



Terminal Tools

VIB Data Core Compute Cluster

- Module lister

Files ▾Utilities ▾Interactive Apps ▾My Interactive Sessions

Develop ▾Help ▾Logged in as

Modules Lister

This page lists the **installed software modules** available on the VIB Data Core Compute Cluster via the [Lmod module system](#). Modules are grouped by software tool and list available versions. Information about the modules listed below are provided by [EasyBuild](#), the framework we use to install modules.

If you'd like a specific module installed, you can [contact our support team](#). Alternatively, you're welcome to [install your own local modules](#) following our documentation.

Show entries

Search:

Module	Category	Versions	Description
Abseil	lib	20230125.2-GCCcore-12.2.0  20230125.3-GCCcore-12.3.0  20240116.1-GCCcore-13.2.0  20240722.0-GCCcore-13.3.0 	Abseil is an open-source collection of C++ library code designed to augment the C++ standard library. The Abseil library code is collected from Google's own C++ code base, has been extensively tested and used in production, and is the same code we depend on in our daily coding lives.
absl-py	tools	2.1.0-GCCcore-13.3.0 	absl-py is a collection of Python library code for building Python applications. The code is collected from Google's own Python code base, and has been extensively tested and used in production.
ACTC	lib	1.1-GCCcore-11.3.0 	ACTC converts independent triangles into triangle strips or fans.
adjustText	tools	0.7.3-foss-2021b 	A small library for automatic adjustment of text position in matplotlib plots to minimize overlaps.
aiohttp	lib	3.10.10-GCCcore-13.3.0  3.8.1-GCCcore-11.2.0  3.8.3-GCCcore-11.3.0  3.8.5-GCCcore-12.2.0  3.8.5-GCCcore-12.3.0 	Asynchronous HTTP client/server framework for asyncio and Python.
almostthere	tools	2.0.2-GCCcore-10.2.0 	Progress indicator C library. ATHR is a simple yet powerful progress indicator library that works on Windows, Linux, and macOS. It is non-blocking as the progress update is done via a dedicated, lightweight thread, as to not impair the performance of the calling program.



HPC

Submitting Jobs

Shell approach

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
 - Different names
 - Same functionality

Resource managers

- Define resources
 - same-same, not the same !

```
--partition = <cluster_name>  
--mem = <number>  
--time = hh:mm:ss  
--ntasks = <number_cores>
```

- Scripting = Reproducibility

This app will launch a [redacted] for remote development. More info can be found in the [redacted] documentation. (Depending on the version, the tunnel URL might also be used directly to connect to a VS Code server using a browser; but we recommend the [redacted] app for that).

An UGent user might get an error message **You do not have access to this** during the **Microsoft Authentication** step. If this is the case then, you might have to whitelist yourself at **UGent selfservice deviceCodeAuth**. You have to do it only once and you can provide [redacted] as motivation.

Cluster
donphan (interactive/debug) [red arrow]

Cluster
donphan [red arrow]

Time (hours)
4 hours [red arrow]

Number of cores (and default memory) per node
4 cores (13.3 GiB mem) [red arrow]

☐ I would like to receive an email when the session starts

☐ Show advanced options

Launch

* The VS Code Tunnel session data for this session can be accessed under the **data root directory**.

Resource managers

Slurm vs Torque

Slurm

Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
 - Different names
 - Same functionality

- Submit a job
 - `sbatch script.sh`

- `qsub script.pbs`

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
 - Different names
 - Same functionality

Slurm

- Submit a job
 - `sbatch script.sh`

- Define resources

- `--job-name = <name>` - - -

- `--output = <name.out>` - - -

Torque

- `qsub script.pbs`

- `--N <job_name>`

- `-o stdout.$PBS_JOBID`

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
 - Different names
 - Same functionality

Slurm

Torque

▪ Submit a job

- `sbatch script.sh`

- `qsub script.pbs`

▪ Define resources

- `--job-name = <name>`

- `--N <job_name>`

- `--output = <name.out>`

- `-o stdout.$PBS_JOBID`

- `--time = hh:mm:ss`

- `-l walltime = hh:mm:ss`

- `--mem = <n>G`

- `memory`

- `-l mem = <n> [MB]`

- `-l pmem = <n> [GB]`

Resource managers

Slurm vs Torque

- #SBATCH
- #PBS

Steps to follow:

- Arguments
 - Different names
 - Same functionality

Slurm

- Submit a job
 - `sbatch script.sh`



Torque

- `qsub script.pbs`



Our first job

- Define resources
 - Define variables
 - Think of modules
 - Commands
-
- Run in Slurm and PBS
 - UGent
 - VIB (if possible)

```
#!/bin/bash
#SBATCH --job-name=test_job_submission
#SBATCH --partition=debug_28C_56T_750GB
#SBATCH --mem=8G
#SBATCH --time=00:30:00

# Define variables
DOCKER_IMG="docker://hello-world"

# Practice best practice by purging all loaded modules first
module purge

# Load necessary modules (none needed for this example)
# module load Python/3.13.1-GCCcore-14.2.0

# Pull minimal Docker image using Singularity
singularity pull $DOCKER_IMG

# Sleep for 20 seconds so there is time to see the job via `squeue`
echo 'Going to sleep for 20 seconds... ZZZzzzzz....'
sleep 20
```

Resource managers

Slurm vs Torque

Slurm

Torque

- #SBATCH
- #PBS

Common tasks:

- Cancel a job
 - scancel <id>
 - qdel <id>
- List Jobs
 - qstat
 - squeue

- Submit a job
 - sbatch script.sh
- Cancel a job
 - scancel <id>
- List jobs
 - squeue

- qsub script.pbs
- qdel
- qstat

Resource managers

Slurm vs Torque

Slurm

Torque

- #SBATCH
- #PBS

Common tasks:

- Cancel a job
 - Scancel <id>
 - qdel <id>
- List Jobs
 - qstat
 - squeue

seff <jobID>

Slurm_jobinfo <jobID>

- Submit a job
 - sbatch script.sh
- Cancel a job
 - scancel <id>
- List jobs
 - squeue
 - Status possible:
 - R: running*
 - CD: completed*
 - S: suspended
 - PD: pending
 - CG: completing

- qsub script.pbs
- qdel
- qstat
- Status possible:
 - R: running*
 - C: completed*
 - S: suspended
 - Q: queued
 - E: exiting

HPC

Connect cmd-line

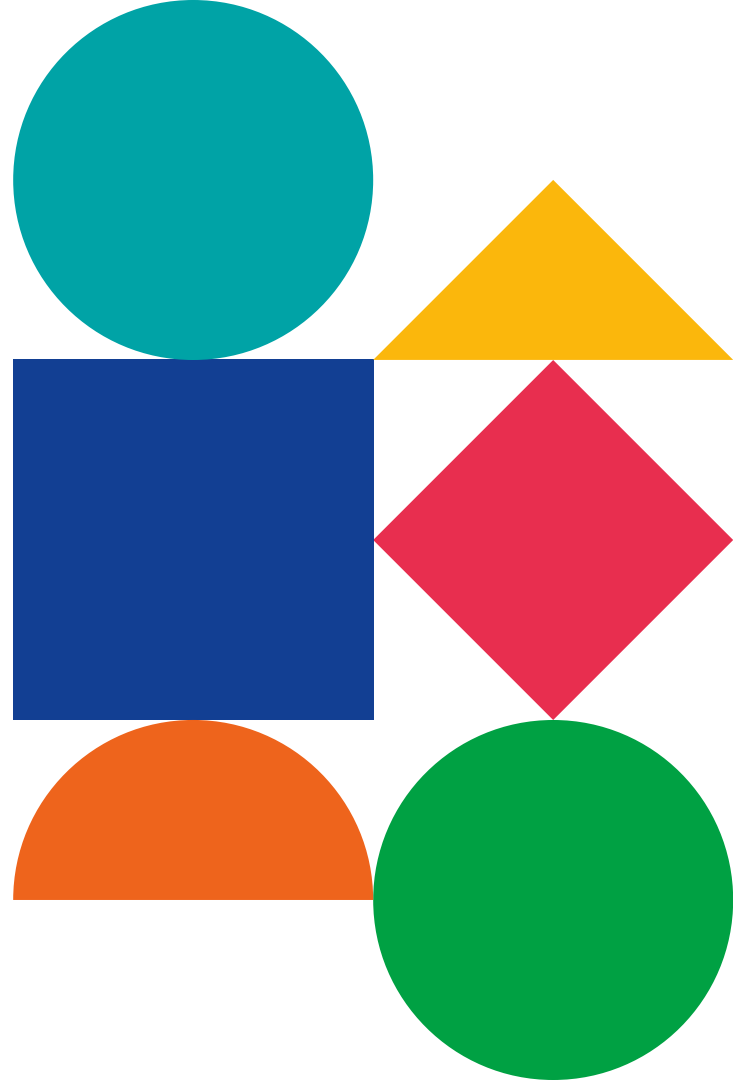
Access from WSL/LINUX terminal

- Log in
 - `ssh vsc00000@login.hpc.kuleuven.be`
 - `ssh vsc47776@login.hpc.ugent.be`
 - `ssh -p 2022 firstname.lastname@compute.vib.be`

It can take some time to be active

Bruna Piereck

bruna.piereckmoura@vib.be



Primary color palette

Aqua
HEX #02A2A7
RBG 2, 162, 167

Orange
HEX #EC6523
RBG 236, 101, 35

Purple
HEX #693A91
RBG 105, 58, 145

Navy
HEX #14274C
RBG 20, 39, 76

Secondary color palette

Blue
HEX #1F448F
RBG 31, 68, 143

Cherry
HEX #E83050
RBG 232, 48, 80

Green
HEX #00A14A
RBG 0, 161, 74

Yellow
HEX #F9B418
RBG 249, 180, 24

Tint 01
HEX #29B3B8
RBG 41, 179, 184

Tint 01
HEX #EF7739
RBG 239, 119, 57

Tint 01
HEX #784B9A
RBG 120, 75, 154

Tint 01
HEX #24539F
RBG 36, 83, 159

Tint 01
HEX #EC4D6A
RBG 236, 77, 106

Tint 01
HEX #1AAF5E
RBG 26, 175, 94

Tint 01
HEX #FCC747
RBG 252, 199, 71

Tint 02
HEX #99D7D6
RBG 153, 215, 214

Tint 02
HEX #F7C1A3
RBG 247, 193, 163

Tint 02
HEX #C2AFD0
RBG 194, 175, 208

Tint 02
HEX #A1B2D5
RBG 161, 178, 213

Tint 02
HEX #F6AAB9
RBG 246, 170, 185

Tint 02
HEX #9BD3B3
RBG 155, 211, 179

Tint 02
HEX #FCE39B
RBG 252, 227, 155

Tint 03
HEX #CBEAEC
RBG 203, 234, 236

Tint 03
HEX #FBE8DF
RBG 251, 232, 223

Tint 03
HEX #E3DCEA
RBG 227, 220, 234

Tint 03
HEX #D5DBEC
RBG 213, 219, 236

Tint 03
HEX #FCE3E6
RBG 252, 227, 230

Tint 03
HEX #D1EAD9
RBG 209, 234, 217

Tint 03
HEX #FEF4CF
RBG 252, 244, 207